

Detailed Summary Report

Objective

The objective of this week task is to learn Python basics and perform basic data exploration and cleaning using the Pandas library. The task involves -

- Loading a CSV dataset
- Exploring the structure
- Handling missing values
- Performing filtering and selection operations
- Removing duplicate records
- Creating derived columns
- Saving the cleaned dataset

Steps Performed

- Loaded the CSV dataset into a Pandas DataFrame using `pd.read_csv()`.
- Explored the dataset using `head()`, `tail()`, `shape`, `columns`, and `dtypes`.
- Identified missing values using `isnull().sum()` and handled them using `fillna()` or `dropna()`.
- Performed basic operations such as filtering rows and selecting columns.
- Removed duplicate rows using `drop_duplicates()`.
- Created a derived column such as `total_amount = initial_price * rating`.
- Saved the cleaned dataset into a new CSV file using `to_csv()`.

Conclusion

This task helped in understanding the basics of Python and Pandas for data analysis and preprocessing.